

Solution Requirements

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements(FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements(NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1.	Authentication	Users authenticate via email/password or OAuth 2.0 (Google Single Sign-On). Sessions are secured using JSON Web Tokens (JWT) or Streamlit session state credentials	High
2.	Authorization levels	Two user roles: Student User (can access personal quiz engine, view personal retention analytics, and edit profile	High

		settings) and Administrator (can view platform usage metrics, manage content templates, and monitor API quotas).	
3.	External interfaces	Integrates with Google Gemini API for adaptive quiz generation and explanation synthesis. Connects to Streamlit Cloud for web app hosting and deployment	High
4.	Transactions processing	Handles active quiz execution workflows, capturing user inputs, real-time response evaluations, confidence score submissions, and immediate memory decay score calculations.	High
5.	Reporting	Generates real-time visual dashboards featuring retention decay curves, subject mastery breakdown charts, and downloadable summary reports (PDF/CSV).	Medium
6.	Business rules, etc	Exponential forgetting curves determine revision dates. If retention probability drops below 60%, the system flags the subject for immediate review	High
7.	Compliance to laws or regulations	Complies with GDPR and standard data	Medium

		privacy guidelines by enabling account deletion, data minimization, and secure handling of user email identifiers.	
8.	Other	Allows users to manually set custom exam target dates to adjust the spacing frequency of predicted revision schedules.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1.	Performance & Speed	Dashboard charts render fast, and Gemini API quiz generations respond without blocking the user interface.	Page components render in < 1.5 seconds ; API response times complete in < 3.0 seconds .
2.	Scalability	System efficiently handles concurrent user sessions and database reads on Streamlit Cloud	Seamlessly supports up to 1,000 active concurrent user sessions without performance degradation.
3.	Security & Data Privacy	API keys are secured via server environment variables, and user data is protected in transit and at rest.	TLS 1.3 encryption in transit and AES-256 encryption at rest; zero API key exposure to client frontend.
4.	Reliability & Availability	Web application remains accessible for continuous daily	99.5% service uptime per month, excluding scheduled

		study routines and scheduled reviews.	maintenance windows
5.	Usability & Accessibility	Modern, mobile-responsive UI with clear contrast and high visual legibility across device screens.	Adheres to WCAG 2.1 AA standards with intuitive navigation requiring zero training.
6.	Other (Maintainability & Portability)	Modular code structure written in Python/Streamlit with version-controlled Git branches.	100% cloud-portable across any environment running Python 3.10+ and Streamlit runtimes.